

Edge-Guided Flow Matching for Controllable Face Synthesis

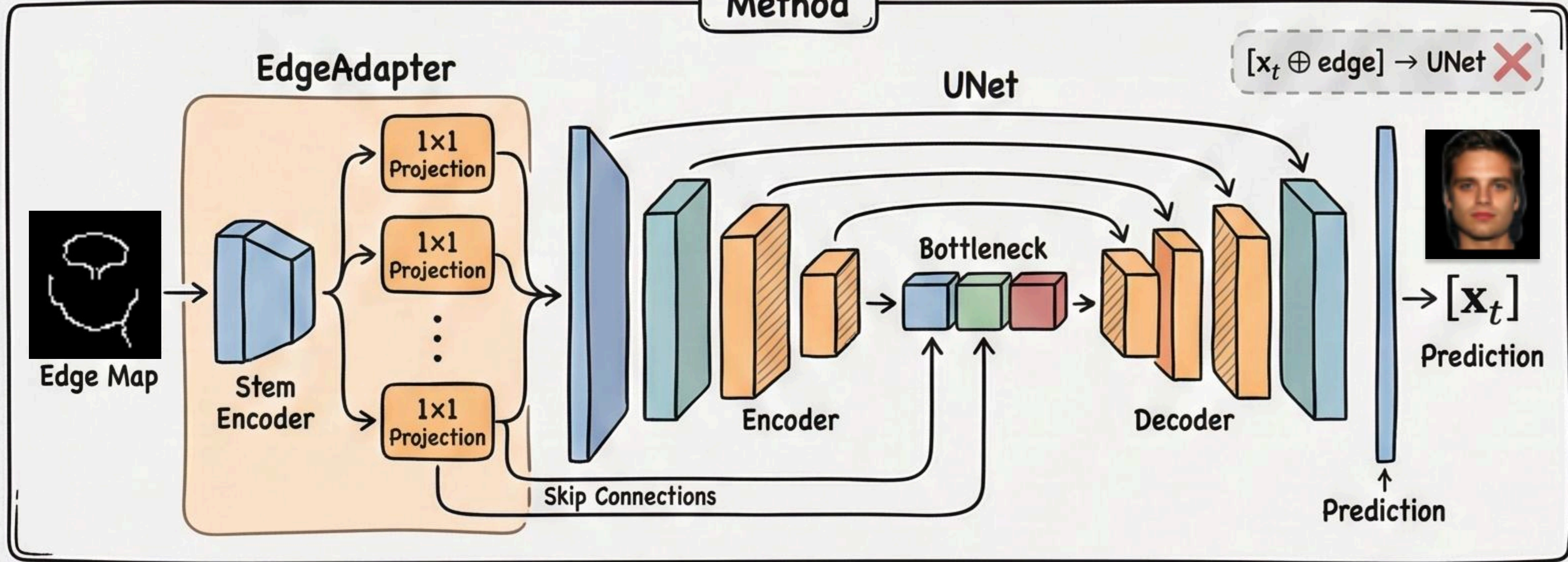
EdgeAdapter Conditioning + Edge-Aware Objectives in Pixel-Space Flow Matching

Sai Ravi Teja Gangavarapu | CMU 10-799: Diffusion & Flow Matching | Spring 2026

Motivation

- We want to generate realistic faces from sparse edge sketches
- Simple channel concat (our HW3 baseline) doesn't enforce structure well

Method



Data

- Canny sigma-mix (multi- σ) for sketch robustness
- Paired geometric augmentation
- Background-removed CelebA

Architecture

- EdgeAdapter: stem encoder $\rightarrow$ per-skip 1x1 projections + middle-block injection
- UNet backbone unchanged (3-ch input)

Training + Inference

- Cycle loss: $\text{Sobel}(\hat{x}_1)$ vs edge map (L1)
- Timestep weighting: scale edge by $(1-t)^\alpha$
- CFG at inference:
$$v = v_{\text{uncond}} + s \cdot (v_{\text{cond}} - v_{\text{uncond}})$$

Quantitative Results

Model	KID $\downarrow$	EAS $\uparrow$	Edge F1 $\uparrow$	CSG $\uparrow$	LPIPS-ref $\downarrow$
Baseline (HW3)	0.1274 ± 0.0013	0.7231 ± 0.1492	0.8623 ± 0.1197	0.5515	0.3364 ± 0.0880
HW4 Final	$[0.1661 \pm 0.0014]$	$[0.7659 \pm 0.1427]$	$[0.9035 \pm 0.1325]$	$[0.6457]$	$[0.1737 \pm 0.0804]$

* CSG = EAS IoU(matched condition) - EAS IoU(mismatched condition). Higher means the model actually uses its conditioning.
Controllability improves across all metrics. KID regresses — we characterize this tradeoff explicitly.

Edge Input

Baseline Output

HW4 Output



Same condition, different model

Future Work

- Isolate data effect: ablation of with vs without background removal and geometric augmentation
- Compare edge representations: XDoG vs Canny vs PiDiNet conditioning
- Per-objective ablation: disable cycle loss / contrastive / CFG individually to attribute the controllability gains

Ablation: Effect of LPIPS Loss

Config	KID $\downarrow$	EAS $\uparrow$	CSG $\uparrow$	LPIPS-ref $\downarrow$
+ LPIPS loss	$[0.1329]$	$[0.5900]$	$[0.4723]$	$[0.3777]$
Canny-only (no LPIPS)	0.1174	0.6179	0.5072	0.3821

LPIPS loss slightly narrows the perceptual gap but hurts KID and control metrics — objective trade-offs are not additive.

EdgeAdapter + edge-aware objectives improve controllability across all metrics.
The realism tradeoff (KID) is real and explicitly characterized.